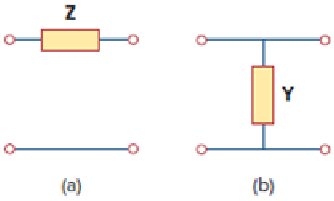
Chapter 19, Problem 43P

Problem

Find the transmission parameters for the single-element two-port networks in Fig. 19.98.

Figure 19.98



Step-by-step solution

Step 1 of 7

Refer to Figure 19.98 in the textbook.

For part (a), redraw the Figure.

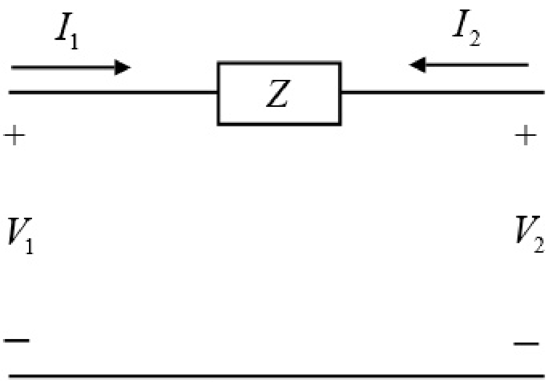


Figure 1

Step 2 of 7

Write the transmission equation.

$$V_1 = AV_2 - BI_2$$
$$I_1 = CV_2 - DI_2$$

For Transmission parameter A,

$$I_2 = 0$$
$$A = \left.\frac{V_1}{V_2}\right|_{I_2=0}$$
$$A = \frac{V_1}{V_1}$$
$$= 1$$

For Transmission parameter B,

$$V_2 = 0$$
$$B = \left.\frac{V_1}{-I_2}\right|_{V_2=0}$$
$$I_2 = -I_1$$
$$A = \frac{V_1}{I_1}$$
$$= \frac{V_1}{\frac{V_1}{Z}}$$
$$= Z\,\Omega.$$

Step 3 of 7

For Transmission parameter C,

$$I_2 = 0$$
$$C = \left.\frac{I_1}{V_2}\right|_{I_2=0}$$
$$I_1 = -I_2$$
$$= 0$$
$$C = \frac{0}{V_2}$$
$$= 0$$

For Transmission parameter D,

$$V_2 = 0$$
$$D = \left.\frac{I_1}{-I_2}\right|_{V_2=0}$$
$$I_1 = -I_2$$
$$D = \frac{1}{1}$$
$$= 1$$

Therefore, the transmission parameters are $\boxed{A = 1, B = Z\,\Omega, C = 0, D = 1}$.

Step 4 of 7

For part (b), redraw the Figure.

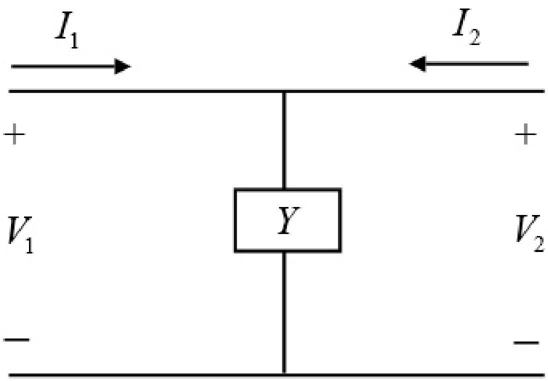


Figure 2

Step 5 of 7

Write the transmission equation.

$$V_1 = AV_2 - BI_2$$
$$I_1 = CV_2 - DI_2$$

For Transmission parameter A,

$$I_2 = 0$$
$$A = \left.\frac{V_1}{V_2}\right|_{I_2=0}$$
$$V_2 = V_1$$
$$A = \frac{V_1}{V_1}$$
$$= 1$$

Step 6 of 7

For Transmission parameter B,

$$V_2 = 0$$
$$B = \left.\frac{V_1}{-I_2}\right|_{V_2=0}$$
$$V_1 = V_2$$
$$B = \frac{0}{-I_2}$$
$$= 0$$

Step 7 of 7

For Transmission parameter C,

$$I_2 = 0$$

$$C = \left.\frac{I_1}{V_2}\right|_{I_2=0}$$

$$V_2 = V_1$$

$$C = \frac{I_1}{V_1}$$
$$= Y \text{ S}$$

For Transmission parameter D,

$$V_2 = 0$$

$$D = \left.\frac{I_1}{-I_2}\right|_{V_2=0}$$

Since, voltage across Y is 0.

$$I_1 = -I_2$$

$$D = \frac{1}{1}$$
$$= 1$$

Therefore, the transmission parameters are $\boxed{A = 1, B = 0\ \Omega, C = Y \text{ S}, D = 1}$.